

GENERAL SPECIFICATION G8.20

Miscellaneous Architectural Works

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G8.20 MISCELLANEOUS ARCHITECTURAL WORKS

G8.20.1 Cabinets

- (a) The Contractor shall provide high and low level cabinets, open shelves, work benches, counters and all other cabinets complete with doors, shelves, accessories, etc. as specified.
- (b) The Contractor shall note that all dimensions shall be carefully checked on Site before fabrication or commencement of work on Site.
- (c) Unless otherwise stated, all external and exposed shelves & surfaces of cabinets shall be finished with 1.0 mm thick approved laminated plastic veneer of colour and pattern to S.O.'s approval.
- (d) All external doors & drawer surfaces are to be provided with post formed bevelled edges.
- (e) Unless otherwise stated all internal surfaces of cabinets shall be pre- laminated with 1.0 mm thick approved laminated plastic veneer to S.O.'s approval.
- (f) All internal shelves shall be finished with 1.0 mm thick approved laminated plastic veneer of colour and pattern to S.O.'s approval and of brands as above described.

G8.20.2 Magnetic Whiteboards

- (a) Magnetic whiteboards shall be tilted at an angle to be approved by the S.O., consisting of 0.65 mm thick magnetic non-glare white colour porcelain enamelled steel sheet fixed to 10 mm thick paper honeycomb on 0.4 mm thick galvanised steel backing sheet screwed to aluminium framing complete with aluminium edge trimming and aluminium pen tray in timber veneer finish and all necessary supports.

G8.20.3 Retractable Projection Screen

- (a) Provide where specified approved retractable projection screen constructed of tearproof fibreglass matt-white screen attached to rigid steel roller encased in heavy gauge steel case with built-in tensioning devices, manually operated and including all necessary wall or ceiling brackets, fixing away from wall at any angle and hooked at bottom to provide a flat taut projection surface. The fixing details are to be approved by the S.O..

G8.20.4 Toilet and Bath Accessories

- (a) All materials furnished under this Section shall be in conformance with specified standards and approved for use in Singapore. All materials and product installation shall comply with local building code requirements.
- (b) As a minimum, the Contractor shall submit the following items:
 - (i) Manufacturer's literature clearly indicating:
 - 1) Identification mark used in the drawings or schedules, size, and description of components;

- 2) Two Samples (50 mm x 100 mm each) of base material with surface finish inside and out;
 - 3) Hardware and locks and attachment devices;
 - 4) Description of rough-in framing; and
 - 5) Details of blocking and anchorage required.
- (ii) Distributor's List: List of local distributors for all supplies required for the accessories installed; and
- (iii) Cleaning instructions.
- (c) Furnish accessory items listed, where indicated by mark or note on Drawings:
- (i) The products and manufacturers listed in the table below have been selected to describe the functional requirements of the toilet and bath accessories and the quality of construction required for this Contract. This list is not all-inclusive; however, all products submitted shall meet the quality and the configuration of the toilet and bath accessories listed.

Item	Mark	Bobrick	Toto	Johnson Suisse	Bradley
Surf. Mounted Roll Toilet Paper Dispenser	TPD	B-2740	TTD-2	BP-202	5241-50
Deck Mounted Liquid Soap Dispenser	SD-1	B-822			6324-68
Mirror and Shelf, Size 1.1 m high x 1.4 m wide	MIR-S	B-292	N.A.	N.A.	7805
Surf. Mounted Paper Towel Dispenser	PTD-2	B-262	P-928	BP-53	250-15
Surf. Mounted Waste Receptacle	WR-1	B-275	WR-1	BW-0015	355
Surf. Mounted Seat Cover dispenser	SCD-2	B-221	1018	N.A.	5831
Mop and Broom Holder (600 mm)	M&BH	B-223 x 24	MS-1	N.A.	9953
Robe Hook	RH	B-6727	TS118W S	BH-06	9124
Grab Bars (toilet, corner type)	GB-1	B-5837	N.A.	N.A.	832-059

- (ii) Finishes shall be satin stainless steel. Manufacturer's or brand name on face of units is not acceptable.
- (iii) Contractor to provide anchors, fasteners, or other devices necessary for a complete, secure installation. Fasteners to consist of tamper-proof screws or bolts.
- (iv) Contractor shall provide fill supplies, such as paper goods, soap, and napkins, as recommended by accessory manufacturer.

G8.20.5 Mild Steel Metal Work

- (a) All building finishes sections to be made from steel plates, rods and bars.
- (b) All mild steel sections including fasteners shall be hot-dip galvanised. Any damaged cut/welded parts shall be treated with two coats of appropriate primers and zinc based compatible paint or equivalent respectively.
- (c) For non-galvanised mild steel members, they shall be prepared and primed with two coats of red lead primer before delivery to Site.
- (d) All galvanised steel bolts used shall include heads, nuts and washers.

G8.20.6 High Pressure Laminates

- (a) High-pressure laminate finishes shall be provided for all joinery work wherever specified. The laminates shall be of first quality and shall be of patterns and colours selected by the S.O..
 - (i) The high-pressure laminates shall be manufactured in a flat press by a combination of decorative papers saturated in melamine resin with phenolic-impregnated layers at pressures exceeding 1000 psi (6895 kPa) and temperatures approaching 150°C;
 - (ii) Thickness of laminates shall be minimum 1.0 mm; and
 - (iii) The Contractor shall bond all laminates with an approved adhesive used in accordance with the manufacturer's instructions, to a suitable backing sheet, such as high-quality plywood, medium to high density fibreboard, particle board or metals; to minimise warpage.

G8.20.7 Carpentry and Joinery

- (a) Where specified, codes and standards shall be taken as a guide only as minimum in quality and performance. Any materials which do not comply with these requirements or which in the opinion of the S.O. are defective or damaged by exposure or other causes shall be removed from the Site.
- (b) All timber used in the Permanent Works shall be primary hardwood first class Chengal, Merbau, Balau or Resak or as specified on the Drawings and shall be thoroughly seasoned, free from large loose or dead knots, injurious open shakes, sap, decay, insect attack or other defects. Well-seasoned and tanalised Meranti may be used for furniture and wall panels.
- (c) All timber shall be graded in accordance with the latest approved rule of the Grading Rules for Sawn Hardwood Timber by the Singapore Timber Association (STA) or, if imported by an equivalent body of the originating country, and the Contractor shall obtain and submit to the S.O. a Grading Certificate from STA. The Contractor shall pay all grading fees, check all grading fees and any other charges in connection therewith. The Contractor shall make all necessary arrangements with STA for grading to be carried out. Such fees and charges shall be included in the rates for carpentry and joinery.
- (d) All graded timber shall be marked with the grade mark, the standard name of the timber and the registered number of the Timber Grader. No ungraded timber shall be brought onto the site or into the Contractor's joinery workshop. S.O. may order the removal of any such ungraded timber from the site.

- (e) Before any payment is made to the Contractor in respect of timber used in the Works, the Grading Certificate covering the said timber must be produced to the S.O..
- (f) The grade of timber used in the works shall not be lower than "Standard & Better Grade M.G.R" This grade allows a small number of defects in the timber, including a limited amount of sapwood. All visible surfaces of timber when fixed in position shall be entirely free from sapwood, knots, and any other defects.
- (g) All timber shall before grading, be properly open stacked under cover until the moisture content has fallen to 18%. The Contractor shall determine the moisture content either by a moisture meter or by the standard weighing method.
- (h) After the Contractor has taken delivery of the graded and seasoned timber, it shall be stored under cover and shall be kept perfectly dry before use. The S.O. may inspect the timber which is not so stored, until a further moisture content test had been taken of the timber and shows the moisture content is as specified above.
- (i) Materials for structural timber shall comply with the recommendations of PD 6697:2010, BS EN 1996-3:2.
- (j) Unless otherwise directed, all woodwork exposed to view shall be wrought.
- (k) All secondary hard wood timber shall be treated with approved pressure preservative and all timber shall be primed before delivery to the Site for fixing.
- (l) Plywood shall comprise resin bonded multiple veneer plywood of thickness specified, finished in matched veneers on both sides.
- (m) All plywood shall be sound, free from warping, twisting and shall comply with BS EN 13986 and shall be of the Weather and Boil Proof quality. Teak-veneered plywood to be of an approved manufacture and appearance. Each sheet of plywood delivered to the site shall bear the following markings:
 - (i) Manufacturer's name or identification;
 - (ii) Standard Number;
 - (iii) W.B.P. (Weather and Boil Proof);
 - (iv) Grade of Finish; and
 - (v) Thickness.
- (n) All wood panels shall comply with BS EN 13986. Unless otherwise specified, board faces shall be Grade 1, or Grade S where a clear finish is specified, board backs shall be Grade 2 and bonding quality BR.
- (o) Hardboard shall comply with **BS EN 622-1** and unless otherwise specified shall be tempered hardboard Type TE of not less than 2.5 mm thickness.

G8.20.8 Treatment of Timbers

- (a) Before being incorporated into the Permanent Works, all timber shall receive a preservative treatment.
- (b) All timber which is to be glued, painted or coated, or which will normally be exposed to view internally shall receive an organic solvent type preservative.

- (c) Before ordering any timbers or panels for incorporation into the Permanent Works, the Contractor shall submit for approval, details of the preservative treatments, the locations where application is to be carried out and details of the quality control and testing involved. Untreated timber shall not be delivered to Site for any purpose unless the S.O. has approved the Contractor's proposals for ensuring that untreated timber is not incorporated in the Permanent Works.

G8.20.9 Pressure Impregnation of Preservative

- (a) All timber shall be pressure treated unless specifically stated otherwise on the Drawings.
- (b) Timber described as "pressure-treated" shall be "tanalised" under vacuum and pressure with proprietary wood preservative to the following nett dry salt retention:

<u>Use</u>	<u>Salt Retention</u>
Structural timber and joinery (below ground)	) 11.0 kg/m ³)
Structural timber and joinery for use the interior of buildings (above ground)	) 5.6 kg/m ³ in)
Exterior, such as weather boarding, door and window frames and cills (above ground)	) 8 kg/m ³)

- (c) The Contractor shall produce a certificate from the supplier that the timber has been pressure-treated in accordance with the manufacturer's instructions and showing the quantities and sizes of timber treated and the net retention of dry salt obtained.
- (d) Ends of pressure-treated timber shall be sealed with a proprietary sealer applied in accordance with the manufacturer's instructions.

G8.20.10 Other Treatment of Timbers

- (a) Except for "pressure treated" timbers, all untreated timber frames, etc., abutting concrete, brick or plaster shall receive one coat of approved wood preservative to BS 144 and SS CP1, well brushed in or primed with one coat of aluminium based wood primer before fixing.

G8.20.11 Size Tolerance

- (a) The sizes of sawn timber, except where otherwise specified, shall be within the margin of permissible variation stated hereunder:
 - (i) For widths, depths or thickness not exceeding 76 mm, within 3 mm of the specified size; and
 - (ii) For widths, depths or thicknesses exceeding 76 mm within 13 mm of the specified size.

- (b) The Contractor shall provide at his own cost any necessary blocks, wedges or battens to compensate for irregular surfaces caused by any variation in size of timber hereby permitted.

G8.20.12 Fixings and Fastenings

- (a) Nails for carpentry work or for fixing steel to timber shall be galvanised.
- (b) Screws shall be countersunk for fixing accessories to timber.
- (c) Brass screws shall be chrome plated and countersunk, raised or dome head for fixing sanitary ware and similar accessories.
- (d) Engine bolts shall be hexagonal head for fixing timber to steel.
- (e) Masonry anchors shall be of expanding screw type in lead in pre- drilled or formed holes grouted in cement mortar.
- (f) Explosive powered steel pins into concrete or steel may be used only if specifically approved by the S.O..
- (g) Glue shall be of the synthetic resin type complying with BS 1204.
- (h) Bolts and screws shall be completed with nuts and washers as appropriate.
- (i) The ironmongery to be supplied by the Contractor shall be carefully fitted and fixed with screws of the same material or finish as the part to be fixed. All exposed parts of locks shall match the finish of the furniture. All joinery woodwork or other work which is damaged, spoiled in appearance or weakened, shall be entirely renewed if so ordered by the S.O. at the Contractor's expense.
- (j) Samples of all ironmongery to be used in the Permanent Works shall be submitted to the S.O. for approval and no ironmongery shall be fitted until it has been so approved.

G8.20.13 Workmanship

- (a) All timber shall be properly stacked on site and protected from weather until they are incorporated in the work. They shall be properly stickered and stacked to allow air circulation around all faces. Any materials which deteriorate as the result of storage shall be replaced at the Contractor's expense.

G8.20.14 Carpentry and Joinery Work

- (a) Carpentry and joinery work shall be carried out in accordance with the best trade practice using competent tradesmen, equipment, materials and processes that are adequate for the purpose.
- (b) Timber which is to be permanently incorporated into the work shall not be used for any temporary purpose.
- (c) All timber members shall be new and selected for straightness. Saw cuts to relieve bowed members will not be accepted.
- (d) All joints shall be secured by concealed fixings where possible. All slips, fillets, casings, wedges, blocks, etc, shall be provided where required.

- (e) All carpentry framing shall be jointed as shown or as is most appropriate in the circumstances. The joints shall be designed and constructed so that they will transmit the loads and resist the stresses to which they will be subjected. Unless otherwise stated, all joints shall be secured by means of notching and where necessary strengthened by wood dowels.
- (f) Fitments to be built between walls and/or columns shall be based on measurements of actual completed work on site such that a neat tight junction is achieved.
- (g) All joinery which is built-in shall be first primed or sealed on the faces and edges to be built-in.
- (h) Where sink units or other items are built into the joinery, they shall be properly fitted and fixed prior to leaving the shop.
- (i) All work shall be thoroughly sand papered leaving no machine marks or sand streaks.

G8.20.15 Toilet Cubicles and Partitions

- (a) The Contractor shall as provide shop drawings, samples and technical data to the S.O. for approval. As a minimum, the following items shall be submitted:
 - (i) Drawings showing detail layout of work; and
 - (ii) Manufacturer's literature indicating fabrication, erection, anchoring, jointing, reinforcing, metal thickness, panel construction, colour chart, and colour samples.
- (b) Toilet partitions shall be in phenolic boards and complete with all necessary accessories, approved by the S.O..
- (c) Construction shall comply with the following:
 - (i) Toilet Partitions: Overhead braced, floor supported toilet enclosures with pilasters extending from floor to headrails;
 - (ii) Style: Flush partitions and doors; and
 - (iii) Components:
 - 1) Stainless-steel Panels: Fabricate of two sheets of Type 316L stainless steel, assembled over and cemented under pressure to a core of sound-deadening insulation and edged with a continuous stainless steel locking strip;
 - 2) Doors: 25 mm thick; sheet 22-gauge;
 - 3) Partitions and Screens: 25 mm thick; sheet 20-gauge;
 - 4) Pilasters: 32 mm thick; sheet 20-gauge;
 - 5) Reinforcement: For tapping, 14-gauge. For anchoring, 12-gauge;
 - 6) Headrails: 25 mm x 37 mm by 20-gauge ASTM A167, Type 316L stainless steel;
 - 7) Fittings: Non-corrosive fittings and brackets and ASTM A167, Type 316L stainless steel pilaster shoes;

- 8) Hardware: Non-corrosive concealed gravity hinges, concealed latch, door stop and keeper, coat hook, and bumper;
 - 9) Finish and Colour; and
 - 10) Stainless Steel: Number 4 Satin Finish.
- (d) Toilet Partitions and Urinal Screens compartments and related items shall be erected in a rigid substantial manner, straight and plumb, and with all horizontal lines level
- (e) Installation shall ensure that:
- (i) Clearances at the Wall: 12 mm for panels and 12 mm for pilasters;
 - (ii) Clearance at Vertical Edges of Doors: Uniform from top to bottom not exceeding 5 mm;
 - (iii) Clearances between Pilasters and Panels: 8 mm;
 - (iv) Door: Free of warp and wind;
 - (v) Conceal all drilling, cutting, and fitting to room finish in the finish work; and
 - (vi) Adjusting: Adjust hardware, leave in perfect working order.
- (f) Site touch-up of scratches to finish permitted only if accepted by the S.O..
- (g) Remove protective maskings and clean surfaces, leaving them free from soil and imperfections.

G8.20.16 Bench and Cupboard Fitments

- (a) All cupboard and fitment dimensions shall be determined accurately by the joiner on site. Fitment dimensions as relevant shall be checked on Site.
- (b) All work shall be carried out by an experienced joinery firm. All materials and workmanship shall be of the highest standard.
- (c) All exposed finished shall be varnished with three coats of ICI Timber Tone or equivalent paint unless otherwise specified or directed by the S.O.. Tops of benches and tops of floor-standing shelves shall be finished with selected plastic laminate of 1.2 mm thickness.
- (d) Unless otherwise directed by the S.O., the construction of cupboards and fitments shall be as follows:
 - (i) Framing and supports shall be ex 75 x 25 mm hardwood timber;
 - (ii) Minimum 25 mm thick veneered plywood to be used for tops, bottoms, ends, intermediate vertical partitions and shelves. Exposed edges shall be finished smooth and painted;
 - (iii) Backing of all cupboard shall be of 18 mm thick plywood;
 - (iv) Each cupboard door shall be hung with 2 Nos or as appropriate brass hinges with approved 75 mm 'D' pull handles; and
 - (v) Drawers shall generally have fronts ex 25 mm timber and sides and backs ex 18 mm, all ploughed to receive 9 mm waterproofed plywood at bottom.

- (e) Drawers shall be fitted with approved drawer slide runners with stops and 'D' pull handles, similar in pattern as for cupboard doors.

G8.20.17 Overhead Shelving

- (a) Overhead shelf shall be of 18 mm thick waterproof plywood, unless otherwise shown on the Drawings or directed by the S.O., with 10 mm deep edge all round, properly varnished to approved colour. Each shelf shall be supported on 4 Nos 150 mm x 4 mm thick galvanised mild steel angle brackets concealed behind wall lining. The brackets shall be securely fixed to the wall and to the under side of the shelf.

G8.20.18 Sheet Metal Flashing and Trim

- (a) Work Included under this Section includes, but is not limited to, Sheet Metal Roofing Works, the provision of the following:
 - (i) Coping, parapet, and cap flashings;
 - (ii) Fascias;
 - (iii) Roof flashings;
 - (iv) Counterflashings over membrane base flashings;
 - (v) Roof joint cover flashings;
 - (vi) Counterflashings at roof-mounted mechanical equipment and vent stacks;
 - (vii) Counterflashings for roof hatches;
 - (viii) Sheet metal roofing, fascias, scuppers and related flashing;
 - (ix) Connection hardware for attachment of panel sheets to building structure;
 - (x) Sealants between metal panel roofing components and between adjoining constructions; sealants around penetrations in the roofing and copings and between copings and flashings;
 - (xi) Drilling and tapping of structure as required for fastening of all work included in this Section;
 - (xii) Cutting and flashing required for roof penetrations;
 - (xiii) Roof insulation and vapour barrier below metal roofing; and
 - (xiv) All other Metal Roof System Work described in the Contract Documents and Drawings including all sundry and relevant related works as required by the Architect.
- (b) 10 years warranty shall be in effect from contract completion.
- (c) Materials shall be delivered, handled and stored by methods approved by the manufacturers.
- (d) Care shall be exercised in handling and placing the metal panel roofing sheets to prevent damage likely to impair the adequacy or appearance of the material in the finished structure. Prevent contact with materials during storage which may cause discolouration, staining or damage.

- (e) The Contractor shall provide shop drawings describing material profile, jointing pattern, jointing details, fastening methods, and installation details within 6 weeks of award of Contract. 300-mm-long samples for each material and finish required shall be submitted.
- (f) Roof Flashing and Trim shall be Aluminium Sheeting to ASTM B209, alloy and temper as required for application and finish; 0.8 mm thick; clear anodized finish.
- (g) Accessories shall include:
 - (i) Fastener: galvanized steel, aluminium, or stainless steel with soft neoprene washers at exposed fasteners;
 - (ii) Underlayment: ASTM D226/D226M; No. 30 asphalt-saturated roofing felt;
 - (iii) Protective Backing: bituminous;
 - (iv) Slip Sheet: rosin-sized building paper;
 - (v) Sealant: Joint Sealers for required material and exposure conditions;
 - (vi) Bedding Compound: rubber asphalt or butyl type recommended by manufacturer;
 - (vii) Plastic Cement: ASTM D4586/D4586, Type I;
 - (viii) Reglets: surface-mounted or recessed galvanized steel as indicated on Drawings or otherwise required by design;
 - (ix) Solder: ASTM B32; Class A, Grade 1; type suitable for application and material being soldered;
 - (x) Flux: U.S. Federal Specification A-A-51145D, Type 1, Form A or B;
 - (xi) Sealer Tape: polyisobutylene sealer tape specifically manufactured for setting flanges or bituminous roofing; and
 - (xii) Isolation Tape: butyl or polyisobutylene, internally reinforced, or 0.5-mm-thick minimum polyester.
- (h) Finishes shall comprise:
 - (i) Backpaint concealed metal surfaces with protective backing paint to a minimum dry film thickness of 0.38 mm;
 - (ii) Class I Natural Anodized Finish: AAMA 611-14, AA-M12C22A41; clear anodic coating not less than 0.7 mils (0.018 m) thick;
 - (iii) Primer Coat: Finish concealed side of metal sheets with primer compatible with finish system, as recommended by finish system manufacturer; and
 - (iv) Fluoropolymer Coat Exposed Surfaces: manufacturer's standard two-coat, thermocured, full-strength 70 percent Kynar 500 or Hylar 5000 coating consisting of a primer and a minimum 0.75-mil dry film thickness with a total minimum dry film thickness of 0.9 mil and 30 percent reflective gloss when tested in accordance with ASTM D523. Color to match adjacent wall panels.

- (i) Exposed surfaces of flashing and sheet metal work shall be free of dents, scratches, abrasions, or other visible defects and shall be clean and ready for painting where applicable.

G8.20.19 Metal Canopy

- (a) Work Included under this Section includes, but is not limited to, metal roofing and roofing systems.
- (b) Performance Criteria
 - (i) Roofing system and associated works will remain intact and waterproof under the local or regional ambient climatic conditions; providing adequate means of dealing with vapour pressure, condensation, corrosion and thermal movement; supports the specified imposed loads and types of roof access without impairment of performance; provides the minimum thermal resistance required by the drawings or current legislation in Singapore, whichever is the higher; and satisfies other specified performance requirements.
 - (ii) Roof fall arrest system to BS EN 363. Metal Roof sheeting and fastenings to AS 1562.1 for resistance to load and to wind pressure and design rainfall capacity to cater to PUB requirement.
 - (iii) Roof to be minimum 1 in 80 and ensure there is no ponding on completed roofs and gutters.
 - (iv) To Submit calculations and drawings from a Singapore licensed Professional Engineer concurrently with the shop drawings and showing, but not limited to compliance with all relevant Singapore legislation and regulations.
 - (v) To comply to SS 370 and SS CP 89. Design and installation to comply to AS 1562.1
 - (vi) Material to be zinc coated sheet steel, with insulation, and to be in polyvinylidene difluoride (PVDF) finish to SO's approval.

END OF SECTION